

Lecture Notes IV

Course: General Topology

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Topic: Compactness

Lemma. If $\bigcup \mathcal{D} \supseteq A$ and $\mathcal{D}_A = \mathcal{D} \cap A \equiv \{D \cap A : D \in \mathcal{D}\}$, then $\bigcup \mathcal{D}_A = A$.

Proof: Let $E \in \mathcal{D}_A$ then $E = D \cap A$ for some $D \in \mathcal{D}$ so $E \subseteq A$ and thus $\bigcup \mathcal{D}_A \subseteq A$. Conversely, if we take $a \in A$, then since $\bigcup \mathcal{D} \supseteq A$, then $a \in D$ for some $D \in \mathcal{D}$ so that $a \in D \cap A$ and thus $A \subseteq \bigcup \mathcal{D}_A$.

Proof 4.21 (Compactness criterion for subspaces). Suppose A is compact, then for any open cover $\mathcal{D}$ of A we have that $\mathcal{D}_A \equiv \mathcal{D} \cap A$ is a relatively open cover of A , i.e., a collection of open subsets in A such that their union is A , and since A is compact, there exists a finite subcover $\mathcal{D}'_A$ so that $\mathcal{D}' := \{D : D \cap A \in \mathcal{D}'_A\}$ is a finite subcover of $\mathcal{D}$.

Conversely, suppose every open cover of A has a finite subcover. Then take any relatively open cover of A , then each element in this collection is of the form $V \cap A$ for some V open in X , and thus define $\mathcal{U}$ to be the collection of all such V , then clearly $\mathcal{U}$ is an open cover of A , so that it has a finite subcover $\mathcal{U}'$ and hence $\mathcal{U}' \cap A$ is a finite subcover of the original relatively open cover of A .

Proof 4.22 (The image of convergent sequences along with the convergence points form compact sets). Suppose $\mathcal{U}$ is an open cover of A , then there is some $U \in \mathcal{U}$ such that $x \in U$ and by the convergence criterion, U must contain x_n all but finitely many n , and each of these have an associated $U_{n_i} \in \mathcal{U}$ for which $x_{n_i} \in U_{n_i}$, so that $\mathcal{U}' = U \cup U_{n_1} \cup U_{n_2} \cup \dots \cup U_{n_k}$ is a finite subcover of $\mathcal{U}$.

Proof 4.23 (Main theorem on compactness). Let $\mathcal{U}$ be an open cover of $f(X)$, then for each $U \in \mathcal{U}$, $f^{-1}(U)$ is open in X . Since $\mathcal{U}$ covers $f(X)$, each $x \in X$ is in some $f^{-1}(U)$, so the collection $\mathcal{D} = \{f^{-1}(U) : U \in \mathcal{U}\}$ is an open cover of X , but since X is compact, there is a finite subcover $\mathcal{D}'$ of $\mathcal{D}$ so that $\mathcal{U}' = \{U \subseteq Y : f^{-1}(U) \in \mathcal{D}'\}$ is a finite subcover of $\mathcal{U}$, completing the proof.

Proof 4.25 (Compact subsets can be separated by open subsets). First, if $B = \{q\}$, then for each $p \in A$, there exists disjoint open subsets U_p, V_p that contain p, q , respectively (because X is Hausdorff), and thus the collection $\mathcal{U} = \{U_p : p \in A\}$ is an open cover of A so it has a finite subcover $\mathcal{U}'$, and then let $\mathcal{V}' = \{V_p : U_p \in \mathcal{U}'\}$. Let $\mathbb{U} = \bigcup \mathcal{U}'$ and $\mathbb{V} = \bigcap \mathcal{V}'$, then $\mathbb{U}$ and $\mathbb{V}$ are disjoint by construction and $A \subseteq \mathbb{U}$, $\{q\} = B \subseteq \mathbb{V}$. Next, for any compact B , we already proved that for each $q \in B$ there are disjoint $\mathbb{U}_q, \mathbb{V}_q \subseteq X$ (defined as in the first part of the proof) such that $A \subseteq \mathbb{U}_q$ and $q \in \mathbb{V}_q$, but since B is compact, for any given cover, there is a finite subcover, so finitely many $\mathbb{V}_q$'s cover B , say $\mathcal{V}$, so that letting $\mathcal{U} = \{\mathbb{U}_q : \mathbb{V}_q \in \mathcal{V}\}$ we get that $U = \bigcap \mathcal{U}$ and $V = \bigcup \mathcal{V}$ are the required disjoint open subsets such that $A \subseteq U$ and $B \subseteq V$.

Proof 4.27 (Properties of compact spaces). i) Suppose X is compact and let $A \subseteq X$ be closed, then let $\mathcal{U}$ be an open cover for A so that $\mathcal{U} \cup \{X \setminus A\}$ is an open cover of X , next we take a finite subcover $\mathcal{U}'$ which either contains $X \setminus A$ in the finite collection or not, but in either case, the finite elements different from $X \setminus A$ cover A so by Theorem 4.21 (Compactness Criterion for Subspaces) we conclude that A is compact.

ii) Let X be Hausdorff and $A \subseteq X$ be compact. Then for any point $p \in X \setminus A$ we apply Theorem 4.25 to get disjoint open subsets U containing A and V containing $\{p\}$. In particular, V is a neighbourhood of p contained in A , and since this is for arbitrary p , $X \setminus A$ is open by Theorem 2.7(vii,d) and thus the conclusion now readily follows.

iii) Let X be a metric space (all the following definitions are thus understood in the metric topology), and let A be compact in X , then for any $x \in X$ consider the collection of open balls $\{B_n(x)\}_{n \in \mathbb{N}}$ which is a cover for A , but since A is compact, we can choose a finite subcover $\{B_{n_k}(x)\}$ so that $B_{n_{max}}(x)$ contains A and thus A is bounded.

Proof 4.28 (Heine-Borel on $\mathbb{R}^n$). If $K \subseteq \mathbb{R}^n$ is compact, then by Theorem 4.27 it is closed and bounded. Conversely, if K is closed and bounded, then there is some $R > 0$ such that K is contained in $[-R, R]^n$ (compact for being the product of compacts $[-R, R]$) and hence K is a closed subset of a compact and thus compact, completing the proof.